Q1a.

1. The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 75 Mbps

   2. The bottleneck link is the link with the smallest capacity between R_S, R_C, and R/4. The bottleneck link is R.

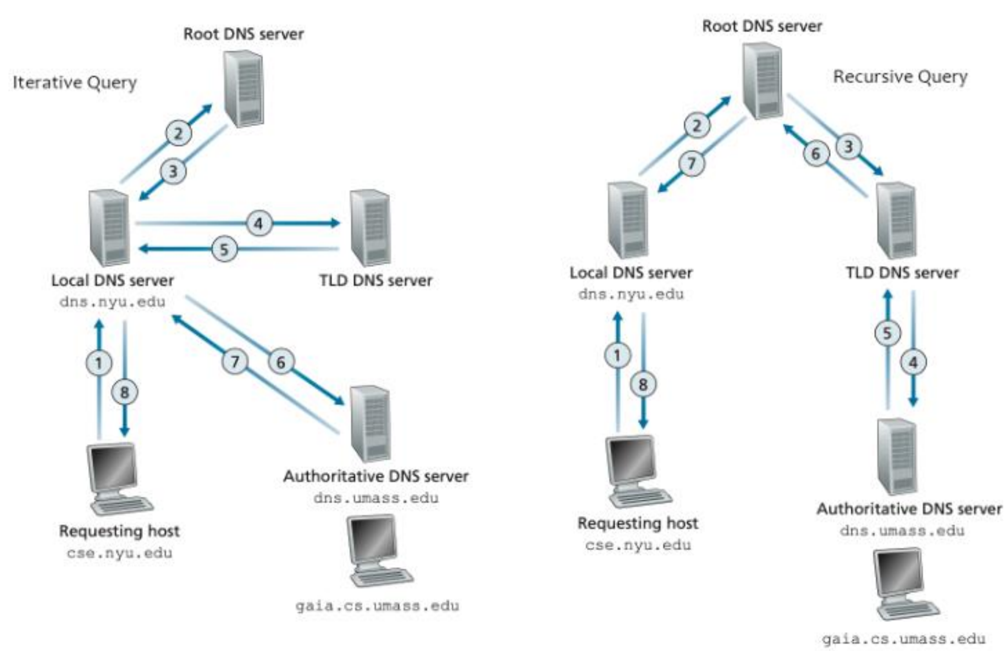
   3. The server's utilization = $R_{bottleneck}$ / R_S = 75 / 90 = 0.83

   4. The client's utilization = $R_{bottleneck}$ / R_C = 75 / 80 = 0.94

   5. The shared link's utilization = $R_{bottleneck}$ / (R / 4) = 75 / (300 / 4) = 1

Q1b.

A recursive DNS lookup is where one DNS server communicates with several other DNS servers to hunt down an IP address and return it to the client. This is in contrast to an iterative DNS query, where the client communicates directly with each DNS server involved in the lookup.



Q1c. 1. (RTT * NUM_PACKETS) + (NUM_PACKETS * (PERCENT__NOT_CACHED / 100) * TRANS_DELAY) = (10 * 80) + (80 * ((100-30) / 100) * 0.25) = 814 ms

Q1d. In a client-server network, we have a specific server and specific clients connected to the server. In a peer-to-peer network, clients are not distinguished; every node act as a client and server. A Client-Server network is more expensive to implement. A Peer-to-Peer is less expensive to implement.

Q2a.

1. DevRTT is calculated with the following equation: (1-beta)*DevRTT + beta * |estimatedRTT - sampleRTT|

   The estimatedRTT for RTT1 is 380

2. EstimatedRTT is calculated with the following equation: (1-alpha)*estimatedRTT + alpha*sampleRTT

   The DevRTT for RTT1 is 65.5

3. TCP timeout is calculated with the following equation: estimatedRTT + (4*DevRTT)

   The timeout for RTT1 is 642

Q2b.

1. Triple duplicate Ack.

2. No, It could be due to reordering, due to queuing and asymmetric path.

3. Timeout.

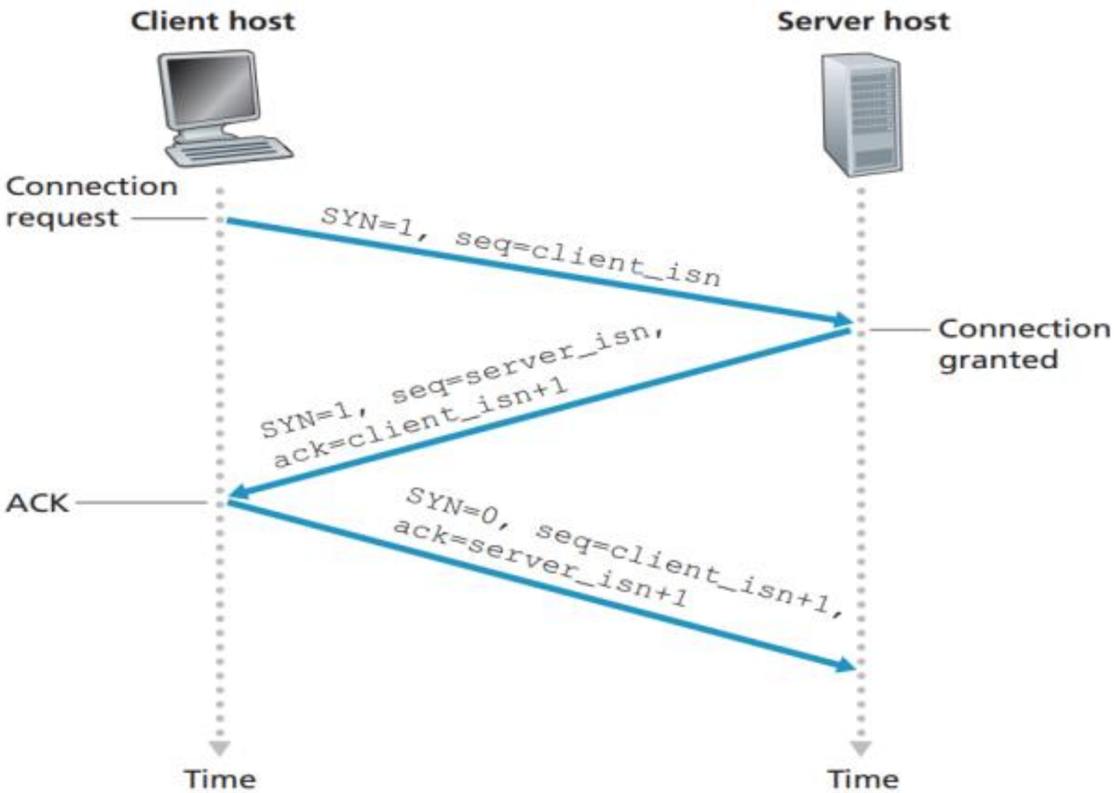
4. No congestion in either direction could cause RTT > RTO (retransmission Time).

5. Slow start is operating during event A.

Q3a

i. Solution:

TCP Three-way handshake



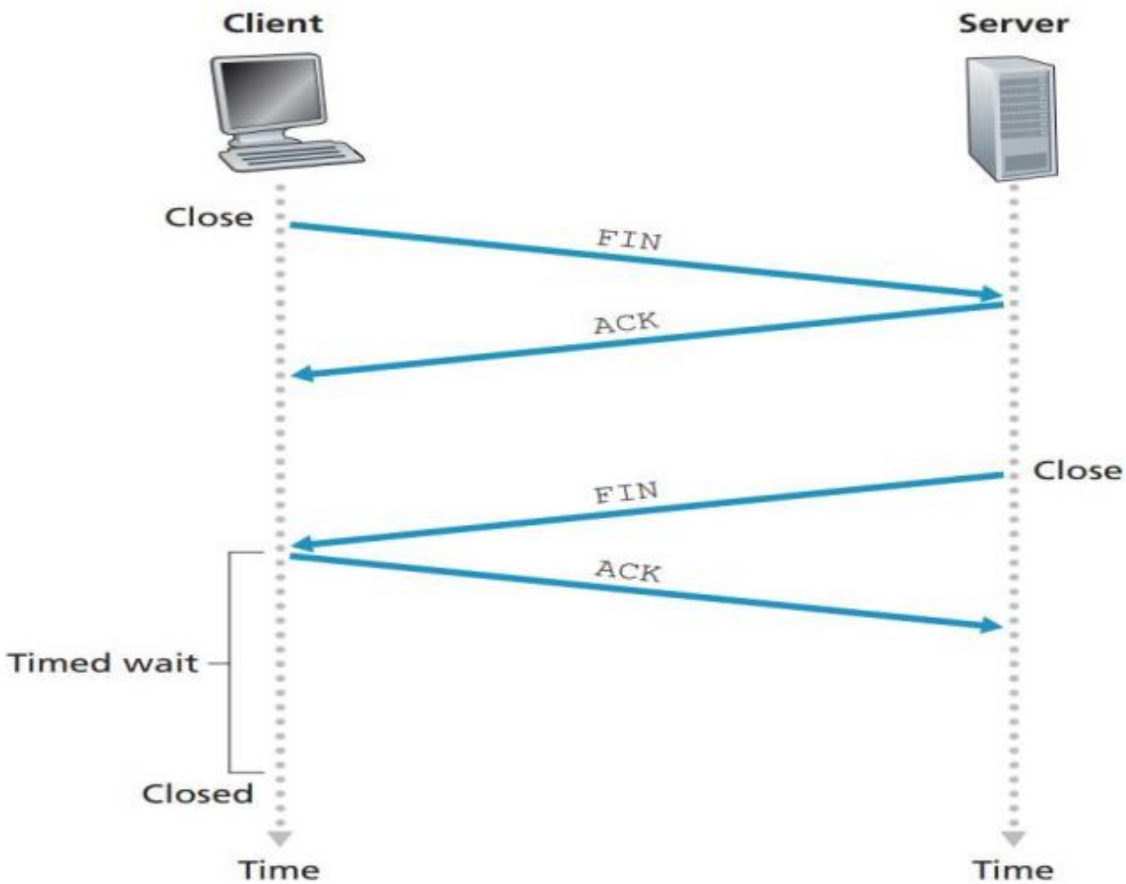
ii. Solution:

Sequence number = 207, source port = 302 and destination port = 80

iii. Acknowledgement number = 127

iv. 1st Acknowledgement number = 207, 2nd Acknowledgement number = 247

V.



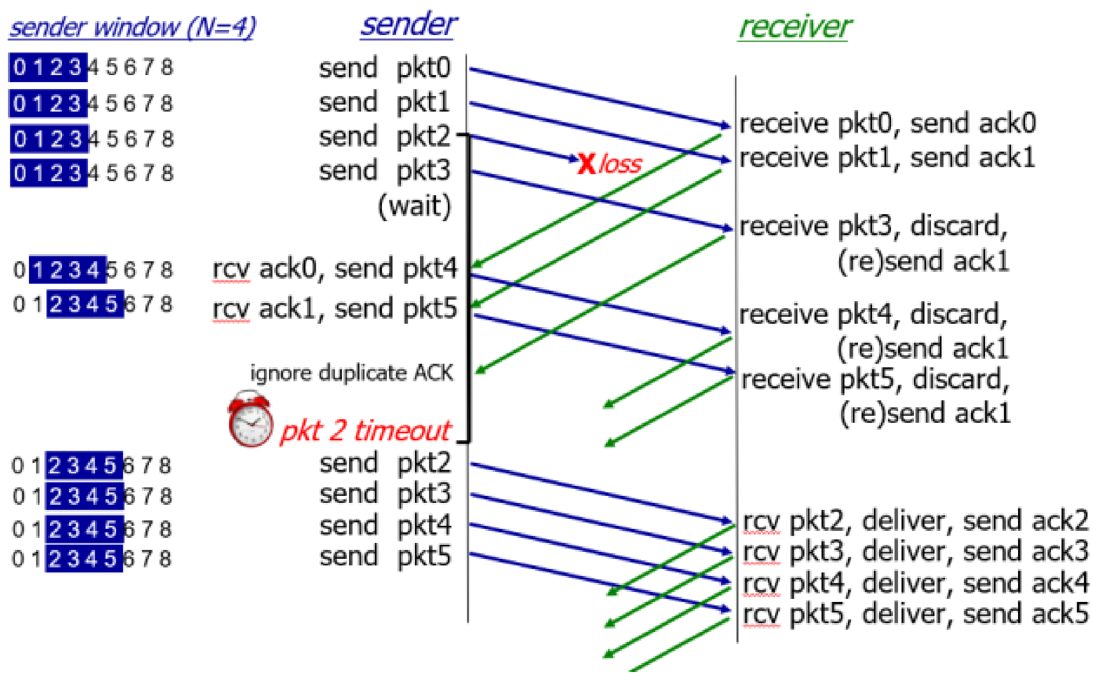
Q3b. IP address, IP address of first hop router, name and IP address of DNS server, Network sub-network mask

Q3c.

Solution: Congestion control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. Congestion control in TCP is based on both open-loop and closed-loop mechanisms. TCP uses a congestion window and a congestion policy that avoid congestion and detect and alleviate congestion after it has occurred. TCP's general policy for handling congestion is based on three phases: slow start, congestion avoidance, and congestion detection.

Q3d. Solution: TCP is considered over UDP for data reliability. Applications which are time tolerant but need data reliability operates over TCP. Examples are HTTP, Email, FTP & etc.

Q3e.



Q4a

Subnet No.	Network Address	Custom Subnet Mask	Usable IP range	Broadcast Address	No. of Hosts
CS	208.16.7.0	255.255.255.128	208.16.7.1 - .126	208.16.7.127	126
SE	208.16.7.128	255.255.255.192	208.16.7.129 - .190	208.16.7.191	62
AI & DS	208.16.7.192	255.255.255.224	208.16.7.193 - .222	208.16.7.223	30
CY	208.16.7.224	255.255.255.240	208.16.7.225 - .238	208.16.7.239	14
Robo	208.16.7.240	255.255.255.248	208.16.7.241 - .246	208.16.7.247	6

Q4b.

1. When the algorithm converges, router X has distance vectors (u,v,w,x,y) = (18,12,11,5,0)

2. The initial distance vectors of router Y are: (u,v,w,x,y) = (x,8,6,0,5) where x is ∞

3. It is called the 'Count to Infinity' problem.

Q4c

i. 2a, 4c, 4a, 3a, 1a and 1b are routers where eBGP is running. AS3 learn about subnet X from AS4 and AS1 AS4,AS2,X and AS1,AS2,X

ii. The client in subnet X will be directed to server B when it wants to access the content. As server B is geographically close to subnet X so, BGP routing protocol feature of IP anycast will choose the closest server. When a BGP router receives multiple route advertisements for this IP address, it treats these advertisements as providing different paths to the same physical location (when, in fact, the advertisements are for different paths to different physical locations). When configuring its routing table, each router will locally use the BGP route-selection algorithm to pick the "best" (for example, closest, as determined by AS-hop counts) route to that IP address.

Q4d

1. The source mac address at point 6 is 06-9A-6C-33-Fo-F7

2. The destination mac address at point 6 is FD-0A-B6-1D-OA-E8

3. The source IP address at point 6 is 128.119.104.178

4. The destination IP address at point 6 is 128.119.196.123